

LG2 Device Specification

Industrial Anchor with PoE Ethernet and WiFi



Overview

The LG1 is an **indoor industrial Anchor with daisy-chained PoE Ethernet and WiFi connectivity** to be used as a Ultra-Wideband Real-Time Location System (RTLS) gateway device managing the communication with the central RTLS server. Its robust design, flexibility and adaptability makes it **ideal for tracking and navigation applications in an industrial environment.**



Figure 1: Front and back views of the LG2 device

Key Features

- Supports both TDoA (Time Difference of Arrival) as well as TWR (Two-Way Ranging) location measurement techniques. Location accuracy sub-0.5m or better.
- Daisy-chained PoE Ethernet backhaul offers an option to reduce infrastructure cabling costs.
- Backhaul via WiFi 2.4GHz b/g/n IoT.
- Supports both UWB channel 5 (6.5 GHz) and channel 9 (8 GHz) with UWB RF compliance compatible with FCC, Japan's ARIB and ETSI of the CE.
- With Low Noise Amplifier, the device provides superior UWB range coverage.
- Integrated Motion sensor: 3-axis accelerometer.
- One multi-function button.
- RGB front LEDs for status indication.
- Buzzer for alarm functions.
- Firmware update over WiFi, Ethernet and USB.

- Free software configuration and visualization tools (for Android, iOS, Windows, macOS and Linux).
- An open [online documentation](#) and [community forum](#).

Electrical Parameters

Parameter	Value
Certification	FCC, CE, ARIB (Japan)
VDC power supply	7VDC to 32VDC, 3W max
USB (power and data)	5VDC 500mA max
PoE	802.3AF, 2.5 max
Operating temperature (without battery) ¹	-40°C - +85°C
UWB supported channels	Channel 5 (6240–6739.2 MHz, CF=6489.6 MHz) Channel 9 (7738–8237.2 MHz, CF=7987.2 MHz, FiRa compatible)
UWB transmit powers	FCC/ARIB/ETSI: -41.3 dBm/MHz max
Bluetooth	2.4 GHz BLE 5.2, TX Power up to +8 dBm
WIFI Frequency Range	2.4 GHz IEEE 802.11 b/g/n
WIFI TX Output Power	19.5 dBm
WIFI Encryption	Open, WPA2-PSK, WPA2-PSK-256, WPA3-SAE
WIFI Country Code	01 (world safe)

Mechanical Parameters

Parameter	Value
Size	W=127 x H=98 x D=41 mm
Weight	135g

Material	ABS
Color	White
IP rating	IP30 (1 screw)
Mounting	Standard 1/4" screwball head camera mount or 4 screws bracket

- Packaging: 1pcs in paper box, 13.5 cm x 10.2 cm x 4.4 cm 137g

About Us

At LEAPS, we specialize in delivering cutting-edge UWB-based localization technology to help businesses and people locate new values.

Our unique technology and innovative solutions created in close cooperation with leading UWB chip producer Qorvo, coupled with a comprehensive range of products, provide unmatched performance and versatility. From tags and anchors to custom-designed systems, our UWB hardware and software is built to meet the specific needs of diverse industries.

Contact Us

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Device Overview

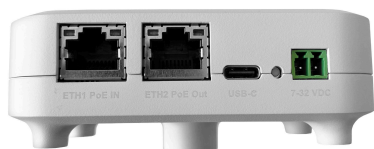


Figure 2: Connectors view of the device

Safety Instructions

- Do not expose devices to water or moisture while in operation.
- Do not expose devices to excessive heat from any source. The product is designed for reliable operation at industrial temperature1.

Warnings

- This product shall be connected only to the USB with a minimum current supply of 0.5 A.
- Any external power supply used with this product shall comply with relevant regulations and standards applicable in the country of intended use.
- This product should be operated in a ventilated environment with a non-condensing environment and should not be covered when being operated.
- There are no user-serviceable parts inside the product, and opening the unit will likely damage the product and invalidate the warranty.
- The cables and connectors of all peripherals used with this product must have adequate insulation to meet relevant safety requirements.

Order Information

- Part number: LR-LG2A1
- Custom code: HS 8517.69.9000